Tejas Jain

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EDUCATION

Qualification	Institution	Year / Batch	Score / Result
B.Tech. in Computer Science Engineering	Graphic Era Deemed to be University, Dehradun, Uttarakhand	**2025–2029**	**9.54 (CGPA - 10 pointer system)**
Class 12 PCM (CBSE)	DPSG Dehradun, Selaqui, Dehradun	**2025**	**96.20 %**
Class 10 (CBSE)	Doon International School, Dehradun	**2023**	**98.16 %**

Anticipated Graduation Date (MM/YY) : 06/29

TECHNICAL SKILLS

- **Programming & Problem Solving :** Python, C, C++, JavaScript, Data Structures and Algorithms
- **Domains:** Machine Learning, Deep Learning, Computer Vision, CNNs, YOLO, Object Detection, Semantic Segmentation, Image Classification, Data Augmentation, Reinforcement Learning, Automation
- **Framework & Libraries :** Numpy, Pandas, Matplotlib, ScikitLearn, TensorFlow, Keras, PyTorch, OpenCV
- **Tools & Platforms :** Git, Raspberry Pi, CARLA Simulator

PROJECTS

Self-Driving Car (CARLA Simulator) |Project Video|GITHUB|

- Demonstrated feasibility of **camera-only perception based** autonomous driving.
- Using a **pretrained segmentation model** detected **drivable area.** Extracted **waypoint arrow overlays** using OpenCV for **navigational cues.** Engineered a **multi-modal steering fusion approach** that utilizes vector analysis to calculate **weighted angular offsets** from **road-mask centroids**, **navigational waypoints**, and **obstacle bounding boxes.** Estimated **object distances** using **YOLO detected bounding boxes** and **dynamically adjusted throttle**.
- **Optimized real-time responsiveness** by **6–10× (boosting inference from 3-4 FPS to 20 - 40 FPS)** by migrating the object detection backbone from **SSD-MobileNet-v2** to **YOLO**. Improvised autonomous driving by refining weighted contributions.

Raspberry Pi Autonomous RC Car |Project Video||GITHUB|

- **Built a Raspberry Pi Autonomous RC Car,** where I **collected data by driving in a simulator** and then **trained a Behavioral Cloning CNN on my driving data.** I then deployed this model on a **differential drive chassis.**
- **Working: Camera captures images** in real time and sends to RPI -> Image sent to CNN -> **CNN makes decision** to steer by a certain angle -> I make use of pigpio library to interface with a **motor controller** that **controls the 4 motors of the differential drive** -> Steer the car according by the use of **different velocities of each wheel.**
- Also experimented with a **servo based steering chassis** for fine tuned autonomous driving.
- Furthermore implemented **multi-key driving control** for **manual control** using a background listener (pynput), enabling combined directional inputs (e.g., forward + left/right) and **incremental steering adjustments** with a **neutral 90° reset** for stability in manual drive.
- Developed practical intuition about **PWM tuning, motor controllers, servo control, battery management,** real-world deployment latency**,** and the gap between simulation and physical autonomy**.**

Cassava Disease Detector (Kaggle Competition) | GITHUB

- **Architected an EfficientNet-based image classifier** using **Transfer Learning** to classify cassava leaf diseases across 5 categories— providing a scalable AI solution for a crop that is grown by **~80% of households** across Sub-Saharan Africa.
- Achieved **82% validation accuracy** during training by implementing a **robust pipeline of data augmentation and pre-processing,** though final competition accuracy was much lower**, highlighting challenges in dataset shift and generalization between local validation and the hidden Kaggle test set.**
- **Optimized inference consistency** by experimenting with **Test-Time Augmentation (TTA)** and investigated the discrepancy between local validation metrics and hidden Kaggle test-set performance.
- **Learning:** Developed intuition on **cross-validation strategies, overfitting in high-cardinality datasets**.

ACADEMIC ACHIEVEMENTS

- **IIT Mandi (2025):** Offer received – Integrated MBA (Data Science & AI)
- **NIT Jamshedpur (2025):** Offer received – B.Tech (Metallurgical & Materials Engg.) via JEE Main
- **JEE Advanced (2025):** Qualified
- **Class 12 Board Examinations (2025):** Ranked 3rd in School
- **Class 10 Board Examinations (2023):** Ranked 3rd in School